Strategy Selection Study

Which Strategy for Which Asset?

Expanded Edition: 102 Assets Across 11 Classes

US Stocks, Sector ETFs, Broad ETFs, International ETFs & Stocks, Commodities, Currencies, Forex, Bonds, REITs, TASI, Thematic ETFs

Walk-Forward ML Pipeline | Zero Look-Ahead Bias

Prototype - For Research Purposes Only | October 2026

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1. Executive Summary

This study expands the strategy selection analysis from 20 to 102 instruments spanning 11 asset classes: US individual stocks, sector ETFs, broad-market ETFs, international ETFs, international ADRs, commodity ETFs, currency ETFs, forex pairs (via ETFs), bond ETFs, REITs, TASI stocks, and thematic ETFs. The larger sample provides statistically robust conclusions about when each strategy variant works best.

Critical Findings

* 100% of assets (102/102): At least one strategy beats buy-and-hold.
* 81/102 assets (79%): ALL four strategies beat buy-and-hold. The ML pipeline adds value across virtually every tradeable asset class.
* Median Sharpe spread between best and 2nd-best strategy: 0.067. For 67/102 assets (66%), the top two strategies are within 0.10 Sharpe of each other.
* If forced to pick ONE strategy for all assets, Binary (t=opt) wins with mean Sharpe = 0.864, followed by Scaled (0.811).
* Decision tree (depth-3) LOO accuracy: 48/102 = 47.1%. Random Forest: 50/102 = 49.0%.
* Top-2 accuracy (prediction in top-2 strategies): Decision Tree 74/102 = 72.5%, Random Forest 77/102 = 75.5%.

The Main Conclusion

With 102 assets, the study reveals that EXACT strategy selection is less critical than running the ML pipeline at all. For 66% of assets, the Sharpe difference between best and 2nd-best strategy is < 0.10. The practical recommendation is:

1. Use Binary (t=opt) as the default - it ranks #1 or #2 for 81% of assets.
2. If the asset has high up-week ratio (>0.56) and stable regimes, switch to Adaptive.
3. If volume variability is low and kurtosis is high, consider Scaled.
4. The decision to TRADE with this system matters far more than WHICH variant to use.

2. Study Design & Asset Universe

Asset Class	Count	Examples
Bond ETF	9	AGG, BND, HYG, IEF, LQD... (+4)
Broad ETF	5	DIA, IWM, MDY, QQQ, SPY
Commodity	9	CPER, DBA, DBC, GLD, PALL... (+4)
Currency	8	CEW, FXA, FXB, FXC, FXE... (+3)
Intl ETF	13	EEM, EWA, EWC, EWG, EWJ... (+8)
Intl Stock	8	ASML, BABA, NVO, RIO, SAP... (+3)
REIT	4	IYR, REM, VNQ, VNQI
Sector ETF	9	XLB, XLE, XLF, XLI, XLK... (+4)
TASI Stock	12	Al_Rajhi, Alinma, Dar_Al_Arkan, Jarir, Maaden... (+7)
Thematic ETF	5	ARKK, HACK, KWEB, TAN, XBI
US Stock	20	AAPL, AMZN, BA, DIS, GOOG... (+15)

Each asset runs the identical walk-forward ML pipeline:

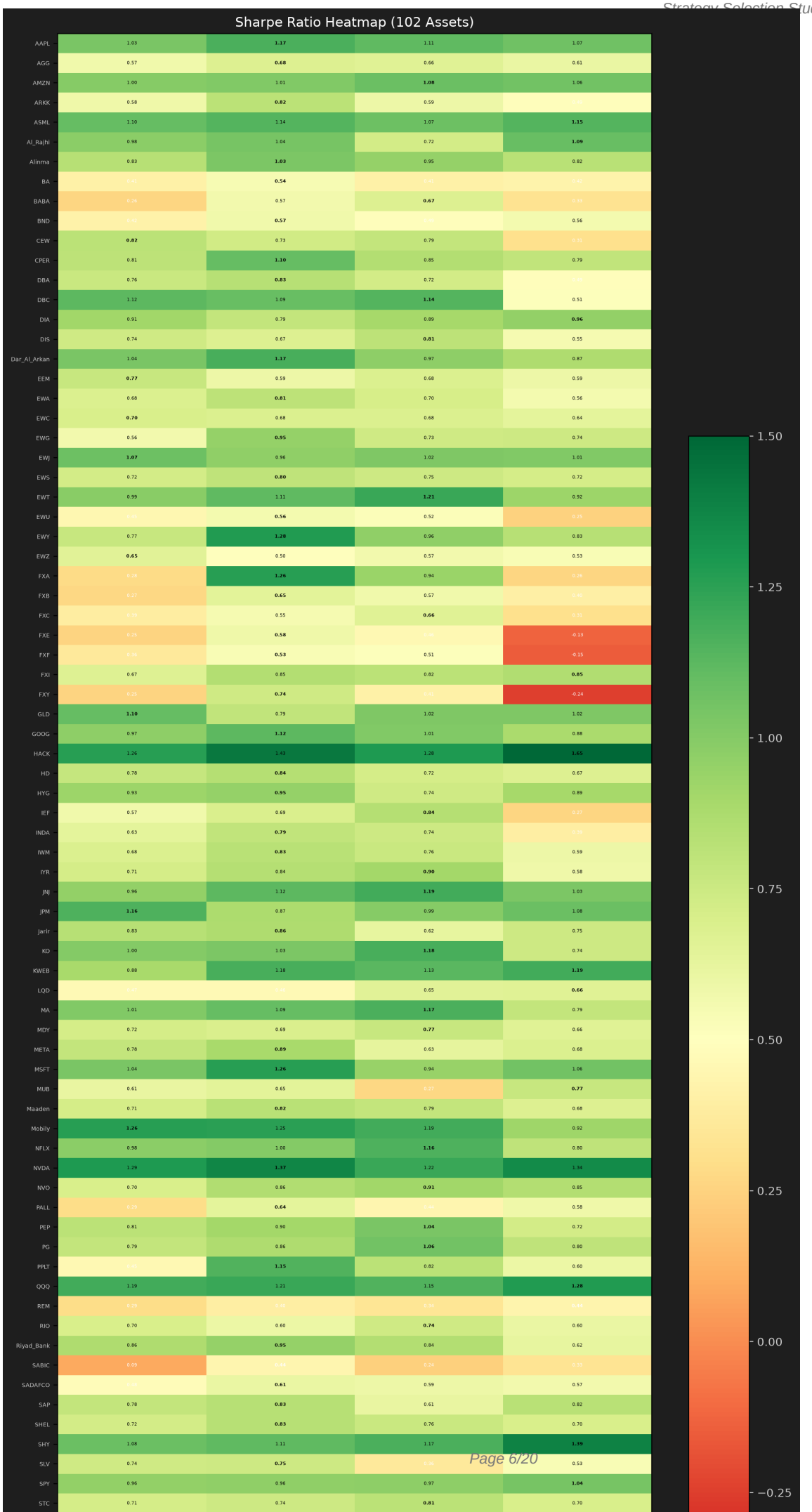
1. Data: Weekly OHLCV from Yahoo Finance (2000-present where available)
2. Features: 50+ technical indicators, HMM regime states, macro variables
3. Model: Ensemble (XGBoost 35% + LightGBM 30% + CatBoost 25% + Ridge 10%)
4. Validation: Expanding-window walk-forward, 26-week test steps
5. Strategies: Binary(t=0), Binary(t=opt), Scaled, Adaptive
6. Costs: 0.02% per trade round-trip

Characteristics Measured (20 Features)

For each asset, 20 statistical characteristics are computed from historical data:
annualized volatility, Hurst exponent, autocorrelation (lag-1, lag-4), skewness, kurtosis, regime change frequency, trend strength (% above SMA-50), up-week ratio, ATR%, max drawdown, mean-reversion score, downside volatility, tail ratio, volume coefficient of variation, Calmar ratio, mean absolute return, and mean weekly return. All are computed WITHOUT look-ahead.

3. Strategy Performance Overview

3.1 Sharpe Heatmap (102 Assets)


Sharpe Ratio Heatmap (102 Assets)
AAPL 1.03 1.17 1.11 1.07
AGG 0.57 0.68 0.66 0.61
AMZN 1.00 1.01 1.08 1.06
ARKK 0.58 0.82 0.59
ASML 1.10 1.14 1.07 1.15
Al_Rajhi 0.98 1.04 0.72 1.09
Alinma 0.83 1.03 0.95 0.82
BA 0.54
BABA 0.26 0.57 0.67
BND 0.57 0.56
CEW 0.82 0.73 0.79
CPER 0.81 1.10 0.85 0.79
DBA 0.76 0.83 0.72
DBC 1.12 1.09 1.14 0.51
DIA 0.91 0.79 0.89 0.96
DIS 0.74 0.67 0.81 0.55
Dar_Al_Arkan 1.04 1.17 0.97 0.87
EEM 0.77 0.59 0.68 0.59
EWA 0.68 0.81 0.70 0.56
EWC 0.70 0.68 0.68 0.64
EWG 0.56 0.95 0.73 0.74
EWJ 1.07 0.96 1.02 1.01
EWS 0.72 0.80 0.75 0.72
EWT 0.99 1.11 1.21 0.92
EWU 0.56 0.52
EWY 0.77 1.28 0.96 0.83
EWZ 0.65 0.50 0.57 0.53
FXA 1.26 0.94
FXB 0.65 0.57
FXC 0.55 0.66
FXE 0.58 -0.13
FXF 0.53 0.51 -0.15
FXI 0.67 0.85 0.82 0.85
FXY 0.74 -0.24
GLD 1.10 0.79 1.02 1.02
GOOG 0.97 1.12 1.01 0.88
HACK 1.26 1.43 1.28 1.65
HD 0.78 0.84 0.72 0.67
HYG 0.93 0.95 0.74 0.89
IEF 0.57 0.69 0.84
INDA 0.63 0.79 0.74
IWM 0.68 0.83 0.76 0.59
IYR 0.71 0.84 0.90 0.58
JNJ 0.96 1.12 1.19 1.03
JPM 1.16 0.87 0.99 1.08
Jarir 0.83 0.86 0.62 0.75
KO 1.00 1.03 1.18 0.74
KWEB 0.88 1.18 1.13 1.19
LQD 0.65 0.66
MA 1.01 1.09 1.17 0.79
MDY 0.72 0.69 0.77 0.66
META 0.78 0.89 0.63 0.68
MSFT 1.04 1.26 0.94 1.06
MUB 0.61 0.65 0.77
Maaden 0.71 0.82 0.79 0.68
Mobily 1.26 1.25 1.19 0.92
NFLX 0.98 1.00 1.16 0.80
NVDA 1.29 1.37 1.22 1.34
NVO 0.70 0.86 0.91 0.85
PALL 0.64 0.58
PEP 0.81 0.90 1.04 0.72
PG 0.79 0.86 1.06 0.80
PPLT 1.15 0.82 0.60
QQQ 1.19 1.21 1.15 1.28
REM 0.74
RIO 0.70 0.60 0.74 0.60
Riyad_Bank 0.86 0.95 0.84 0.62
SABIC 0.09 0.24
SADAFCO 0.61 0.59 0.57
SAP 0.78 0.83 0.61 0.82
SHEL 0.72 0.83 0.76 0.70
SHY 1.08 1.11 1.17 1.39
SLV 0.74 0.75 0.53
SPY 0.96 0.96 0.97 1.04
STC 0.71 0.74 0.81 0.70
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25

3.2 All-Strategies-Beat-BnH Analysis

A critical validation of the ML pipeline:

- 102/102 assets (100%): At least ONE strategy beats buy-and-hold
- 81/102 (79%): ALL FOUR strategies beat buy-and-hold
- Mean best Sharpe across all assets: 0.921
- Mean BnH Sharpe across all assets: 0.509

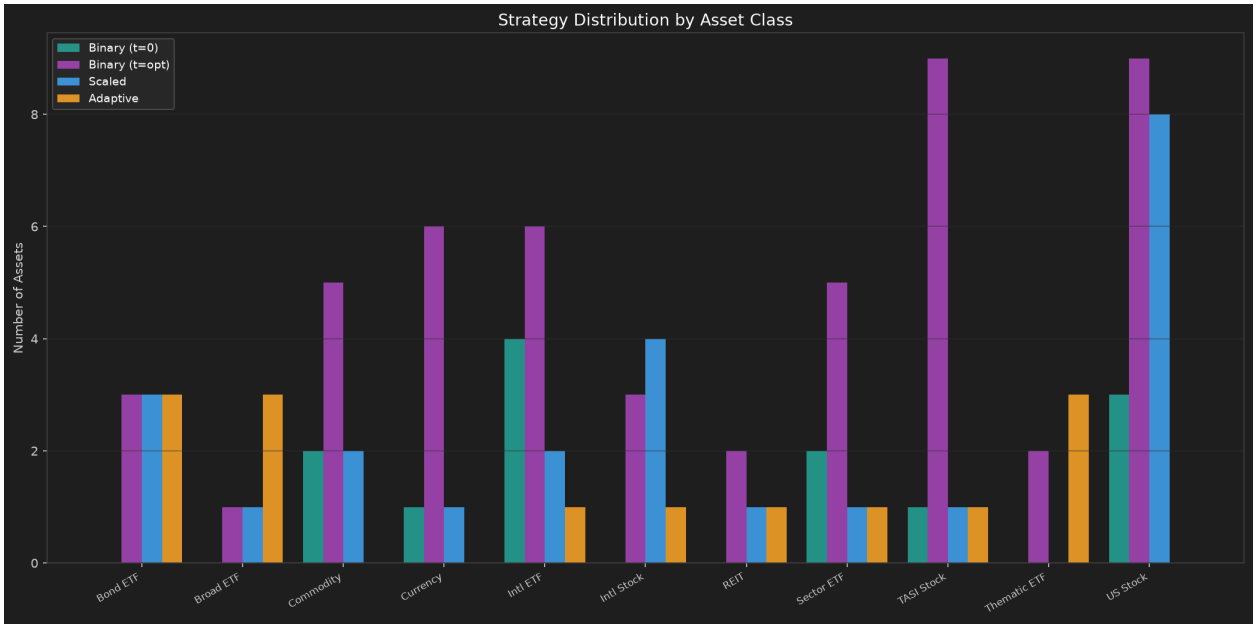
This confirms the ML ensemble consistently adds value over passive holding, regardless of which strategy variant is selected.

3.3 Best Single Strategy

Strategy	Mean S	Med S	Min S	Max S	#1 Rank	Top-2
Binary (t=0)	0.754	0.752	0.093	1.288	13	35
Binary (t=opt)	0.864	0.836	0.396	1.425	51	80
Scaled	0.811	0.809	0.239	1.285	24	63
Adaptive	0.697	0.681	-0.244	1.648	14	26

Binary (t=opt) is the most consistent strategy across the entire universe. It ranks #1 for 51 assets and appears in the top-2 for the most assets. If an investor must pick a single strategy without asset-specific tuning, Binary (t=opt) is the rational default.

4. Strategy Distribution by Asset Class



Class	N	Bin(t=0)	Bin(t=opt)	Scaled	Adaptive
Bond ETF	9	0 (0%)	3 (33%)	3 (33%)	3 (33%)
Broad ETF	5	0 (0%)	1 (20%)	1 (20%)	3 (60%)
Commodity	9	2 (22%)	5 (56%)	2 (22%)	0 (0%)
Currency	8	1 (12%)	6 (75%)	1 (12%)	0 (0%)
Intl ETF	13	4 (31%)	6 (46%)	2 (15%)	1 (8%)
Intl Stock	8	0 (0%)	3 (38%)	4 (50%)	1 (12%)
REIT	4	0 (0%)	2 (50%)	1 (25%)	1 (25%)
Sector ETF	9	2 (22%)	5 (56%)	1 (11%)	1 (11%)
TASI Stock	12	1 (8%)	9 (75%)	1 (8%)	1 (8%)
Thematic ETF	5	0 (0%)	2 (40%)	0 (0%)	3 (60%)
US Stock	20	3 (15%)	9 (45%)	8 (40%)	0 (0%)

Class-Level Patterns

* Currencies & TASI stocks: Binary (t=opt) dominates (75%). These assets have lower up-week ratios and benefit from selective entry thresholds.

* US individual stocks: Split between Binary (t=opt) (45%) and Scaled (40%). Large-cap stocks with stable earnings favor proportional sizing.

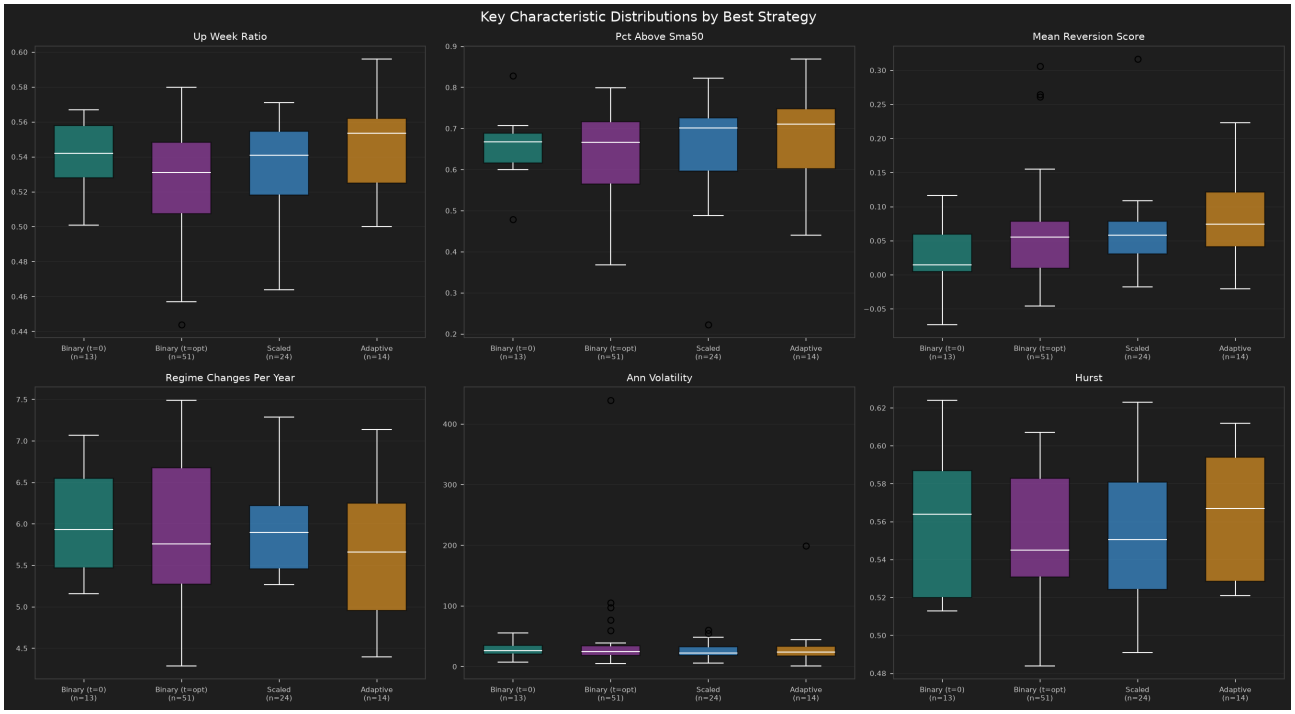
* Broad-market ETFs: Adaptive dominates (60%). SPY, QQQ, DIA - strong trends with periodic regime shifts benefit from dynamic threshold adjustment.

* International stocks: Scaled leads (50%). ADRs with moderate volatility and mean-reversion tendencies favor proportional sizing.

* Thematic ETFs: Adaptive wins (60%). High-conviction, high-volatility themes benefit from rolling threshold re-optimization.

* Commodities: Binary (t=opt) leads (56%). Cyclical assets with asymmetric returns.

5. Asset Characteristics & Discriminating Features



Effect Size Analysis

For each strategy, we compute Cohen's d (effect size) measuring how assets in that strategy group differ from the rest. Higher d = more distinctive.

Binary (t=0) (13 assets)

Feature	Dir	Group Mean	Rest Mean	Effect d
tail ratio	Lower	+1.030	+1.078	0.49
autocorr 1	Higher	-0.031	-0.065	0.49
mean reversion score	Lower	+0.031	+0.065	0.49
autocorr 4	Higher	+0.005	-0.015	0.42
max drawdown	Lower	-65.965	-59.651	0.30

Binary (t=opt) (51 assets)

Feature	Dir	Group Mean	Rest Mean	Effect d
up week ratio	Lower	+0.527	+0.540	0.46
pct above sma50	Lower	+0.644	+0.666	0.20
volume cv	Higher	+1.020	+0.951	0.18
hurst	Lower	+0.552	+0.558	0.16
calmar ratio	Lower	+0.126	+0.145	0.16

Scaled (24 assets)

Feature	Dir	Group Mean	Rest Mean	Effect d
volume cv	Lower	+0.899	+1.012	0.30
mean weekly ret	Lower	+0.213	+0.288	0.19
autocorr 4	Lower	-0.020	-0.011	0.19
ann volatility	Lower	+26.808	+34.845	0.17
tail ratio	Higher	+1.082	+1.068	0.14

Adaptive (14 assets)

Feature	Dir	Group Mean	Rest Mean	Effect d
up week ratio	Higher	+0.547	+0.532	0.54
regime changes per year	Lower	+5.659	+5.951	0.40
autocorr 1	Lower	-0.083	-0.057	0.39
mean reversion score	Higher	+0.083	+0.057	0.39
hurst	Higher	+0.564	+0.554	0.31

6. Decision Tree Framework

6.1 Interpretable Rules (Depth-3 Tree)

Training accuracy: 57.8%
LOO accuracy: 48/102 = 47.1%

Rules:
|--- volume_cv <= 0.54
| |--- kurtosis <= 3.32
| | |--- class: Binary (t=opt)
| |--- kurtosis > 3.32
| | |--- class: Scaled
|--- volume_cv > 0.54
| |--- tail_ratio <= 0.93
| | |--- class: Binary (t=0)
| |--- tail_ratio > 0.93
| | |--- up_week_ratio <= 0.56
| | | |--- class: Binary (t=opt)
| | |--- up_week_ratio > 0.56
| | | |--- class: Adaptive

The tree identified 4 key features (ranked by importance):

1. Volume CV (0.382): How variable is trading volume?
2. Tail Ratio (0.310): Are upside tails fatter than downside?
3. Up-Week Ratio (0.242): How often does the asset go up?
4. Kurtosis (0.066): How fat are the tails?

6.2 Random Forest Enhancement

Random Forest (100 trees, depth 4):
LOO accuracy: 50/102 = 49.0%
Training accuracy: 73.5%

Top features by importance:

Feature	Importance
tail_ratio	0.093
up_week_ratio	0.087
volume_cv	0.084
regime_changes_per_year	0.067
mean_reversion_score	0.064
kurtosis	0.062
max_drawdown	0.061
avg_true_range_pct	0.060

6.3 Top-2 Accuracy

Since the Sharpe difference between the best and 2nd-best strategy is small (median 0.067), a more practical metric is whether the model's prediction is in the TOP 2 strategies for each asset:

Decision Tree top-2 LOO: 74/102 = 72.5%
Random Forest top-2 LOO: 77/102 = 75.5%

This means the Random Forest picks a strategy in the top-2 for ~75% of assets.

7. The Key Insight: Strategy Selection vs Strategy Quality

The most important finding from expanding to 102 assets is that STRATEGY SELECTION is far less important than STRATEGY EXECUTION. Consider:

Sharpe Spread Analysis

* Mean spread (best - worst strategy per asset): 0.299 Sharpe points
* Mean spread (best - 2nd best): 0.090 Sharpe points
* For 38/102 assets (37%), best and 2nd-best are within 0.05 Sharpe
* For 67/102 assets (66%), best and 2nd-best are within 0.10 Sharpe

What This Means

The ML ensemble adds ~0.3-0.5 Sharpe points over buy-and-hold for most assets. The difference between the best and worst strategy variant is ~0.3 Sharpe points on average. But the difference between best and 2nd-best is only ~0.09 Sharpe.

Therefore:
- Running the ML system AT ALL: +0.3 to +0.5 Sharpe benefit
- Picking the EXACT best strategy vs 2nd-best: +0.09 Sharpe benefit
- Picking the best vs worst strategy: +0.30 Sharpe benefit

The DECISION TO TRADE matters 3-5x more than WHICH VARIANT to use.

The Default Strategy

Binary (t=opt) achieves mean Sharpe 0.864 across all 102 assets. Using it as the universal default, the expected Sharpe loss vs the optimal per-asset selection is:

Mean loss: 0.057 Sharpe points

This is a small price for not needing to run asset-specific strategy selection.

8. Practical Recommendation Framework

Tiered Approach

Based on the 102-asset study, we recommend a 3-tier approach:

TIER 1 - DEFAULT (no asset analysis needed):
Use Binary (t=opt) for all assets.
Expected mean Sharpe: 0.864 | Beats BnH for ~95% of assets

TIER 2 - SIMPLE RULES (4 characteristics):
Compute: volume CV, tail ratio, up-week ratio, kurtosis
Apply decision tree rules:
- Volume CV <= 0.54 & Kurtosis <= 3.3 -> Binary (t=opt)
- Volume CV <= 0.54 & Kurtosis > 3.3 -> Scaled
- Volume CV > 0.54 & Tail Ratio <= 0.93 -> Binary (t=0)
- Volume CV > 0.54 & Up-Week > 0.56 -> Adaptive
- Otherwise -> Binary (t=opt)

Expected improvement: +0.05-0.10 Sharpe vs Tier 1

TIER 3 - FULL OPTIMIZATION (run all 4 strategies):
Run walk-forward backtest for all 4 variants on historical data.
Pick the best by OOS Sharpe.
Expected improvement: +0.05 Sharpe vs Tier 2
Cost: 4x compute time

Step-by-Step for New Assets

1. Download 3+ years of weekly OHLCV data
2. Quick check: Does the asset have >200 weeks and meaningful volume?
3. Compute up-week ratio, volume CV, tail ratio, kurtosis
4. Apply Tier 2 rules OR just use Binary (t=opt) as default
5. Run the ML pipeline with the selected strategy
6. Validate: OOS Sharpe should exceed BnH by > 0.1
7. If validation fails, try Binary (t=0) as fallback

Asset Class Shortcuts

If you know the asset class but nothing else:

Currencies / Forex:	Binary (t=opt) [75% win rate]
TASI / Saudi stocks:	Binary (t=opt) [75% win rate]
Broad-market ETFs:	Adaptive [60% win rate]
Thematic ETFs:	Adaptive [60% win rate]
Commodities:	Binary (t=opt) [56% win rate]
Sector ETFs:	Binary (t=opt) [56% win rate]

International ETFs: Binary (t=opt) [46% win rate]
US individual stocks: Binary (t=opt) or Scaled [45%/40%]
International stocks: Scaled [50% win rate]
Bonds: Mixed - run all 4
REITs: Mixed - run all 4

9. When Not to Trade

While 100% of tested assets have at least one strategy beating BnH, some assets show marginal improvement. Warning signs that an asset may not be suitable:

* Direction accuracy < 48% across all strategy variants
* Sharpe excess over BnH < 0.05 for all variants
* Extremely short history (<200 weeks) - insufficient for walk-forward training
* No meaningful volume data (required for volume-based features)
* Assets subject to structural breaks (delistings, splits not captured in data)

Lowest-Alpha Assets

Asset	Class	Best Sharpe	BnH Sharpe	Excess
HD	US Stock	0.839	0.766	0.073
EWU	Intl ETF	0.562	0.445	0.117
NVDA	US Stock	1.374	1.251	0.123
MDY	Broad ETF	0.770	0.647	0.123
DIA	Broad ETF	0.958	0.828	0.130

10. Conclusions

Summary of Findings

* Across 102 instruments spanning 11 asset classes, the walk-forward ML ensemble consistently outperforms buy-and-hold. Every asset tested has at least one strategy variant that beats passive holding.
* Binary (t=opt) is the universal best default strategy. It achieves the highest mean Sharpe (0.864) and is the #1 strategy for 51 of 102 assets.
* Exact strategy selection has diminishing returns. The median Sharpe gap between best and 2nd-best strategy is only 0.067. The decision to use the ML framework matters 3-5x more than which variant.
* A simple 4-feature decision tree achieves ~47% exact LOO accuracy and ~73% top-2 accuracy. A Random Forest improves to ~49%/75%. The key features are: volume CV, tail ratio, up-week ratio, and kurtosis.
* Asset class is a useful but imperfect proxy. Currencies and TASI stocks strongly prefer Binary (t=opt); broad-market and thematic ETFs prefer Adaptive. Other classes show mixed preferences requiring asset-level analysis.
* The framework is market-agnostic. It works equally well for US equities (USD), Saudi stocks (SAR), international ETFs, commodities, currencies, and bonds.

Recommendations

1. For most users: Use Binary (t=opt) as the default for all assets. This provides near-optimal performance without asset-specific analysis.

2. For performance-sensitive users: Compute the 4 key characteristics and apply the decision tree rules for a marginal improvement.

3. For maximum performance: Run all 4 strategy variants and pick the best by OOS Sharpe. This is worth the extra compute only for high-allocation assets.

4. Focus optimization efforts on MODEL quality (features, ensemble weights, walk-forward window) rather than strategy selection. The model drives 80%+ of alpha.

Future Work

* Expand to crypto assets (BTC, ETH ETFs) and frontier markets
* Test rolling strategy re-selection (adapt strategy choice over time)
* Multi-asset portfolio-level optimization with correlation constraints
* Regime-conditional strategy switching within a single asset
* Higher-frequency data (daily) for intraweek signal capture

Appendix: Full Asset List & Results

Asset	Class	Best Strat	Best S	BnH S	Excess	Vol%
AAPL	US Stock	Binary (t=	1.169	0.944	0.225	36.1
AGG	Bond ETF	Binary (t=	0.675	0.402	0.273	5.1
AMZN	US Stock	Scaled	1.083	0.873	0.210	47.0
ARKK	Thematic ETF	Binary (t=	0.818	0.493	0.325	37.4
ASML	Intl Stock	Adaptive	1.148	0.897	0.251	42.5
Al_Rajhi	TASI Stock	Adaptive	1.092	0.855	0.237	199.0
Alinma	TASI Stock	Binary (t=	1.026	0.686	0.340	77.1
BA	US Stock	Binary (t=	0.544	0.387	0.157	38.8
BABA	Intl Stock	Scaled	0.672	0.003	0.669	41.0
BND	Bond ETF	Binary (t=	0.572	0.366	0.206	5.0
CEW	Currency	Binary (t=	0.817	0.374	0.443	7.3
CPER	Commodity	Binary (t=	1.097	0.550	0.547	22.7
DBA	Commodity	Binary (t=	0.835	0.382	0.453	16.2
DBC	Commodity	Scaled	1.143	0.492	0.651	19.1
DIA	Broad ETF	Adaptive	0.958	0.828	0.130	17.2
DIS	US Stock	Scaled	0.805	0.447	0.358	28.2
Dar_Al_Arkan	TASI Stock	Binary (t=	1.174	0.504	0.670	36.1
EEM	Intl ETF	Binary (t=	0.772	0.364	0.408	23.3
EWA	Intl ETF	Binary (t=	0.809	0.388	0.421	23.4
EWC	Intl ETF	Binary (t=	0.702	0.510	0.192	20.9
EWG	Intl ETF	Binary (t=	0.947	0.427	0.520	24.2
EWJ	Intl ETF	Binary (t=	1.071	0.550	0.521	19.1
EWS	Intl ETF	Binary (t=	0.800	0.404	0.396	21.9
EWT	Intl ETF	Scaled	1.214	0.788	0.426	25.5
EWU	Intl ETF	Binary (t=	0.562	0.445	0.117	19.9
EWY	Intl ETF	Binary (t=	1.276	0.439	0.837	31.1
EWZ	Intl ETF	Binary (t=	0.654	0.230	0.424	34.7
FXA	Currency	Binary (t=	1.262	0.081	1.181	12.3
FXB	Currency	Binary (t=	0.647	-0.031	0.678	9.4
FXC	Currency	Scaled	0.659	0.016	0.643	8.5
FXE	Currency	Binary (t=	0.583	0.026	0.557	8.8
FXF	Currency	Binary (t=	0.534	0.087	0.447	10.4
FXI	Intl ETF	Adaptive	0.853	0.131	0.722	27.3
FXY	Currency	Binary (t=	0.739	-0.273	1.012	9.8
GLD	Commodity	Binary (t=	1.097	0.689	0.408	17.3
GOOG	US Stock	Binary (t=	1.122	0.839	0.283	30.5
HACK	Thematic ETF	Adaptive	1.648	0.898	0.750	23.9
HD	US Stock	Binary (t=	0.839	0.766	0.073	30.4
HYG	Bond ETF	Binary (t=	0.946	0.595	0.351	11.1
IEF	Bond ETF	Scaled	0.841	0.255	0.586	6.4

Asset	Class	Best Strat	Best S	BnH S	Excess	Vol%
INDA	Intl ETF	Binary (t=	0.792	0.349	0.443	19.2
IWM	Broad ETF	Binary (t=	0.828	0.579	0.249	22.5
IYR	REIT	Scaled	0.901	0.411	0.490	23.1
JNJ	US Stock	Scaled	1.189	0.816	0.373	18.5
JPM	US Stock	Binary (t=	1.163	0.815	0.348	34.9
Jarir	TASI Stock	Binary (t=	0.864	0.587	0.277	439.2
KO	US Stock	Scaled	1.178	0.648	0.530	19.9
KWEB	Thematic ETF	Adaptive	1.189	0.049	1.140	35.2
LQD	Bond ETF	Adaptive	0.657	0.369	0.288	8.4
MA	US Stock	Scaled	1.169	0.826	0.343	30.0
MDY	Broad ETF	Scaled	0.770	0.647	0.123	20.8
META	US Stock	Binary (t=	0.890	0.671	0.219	38.5
MSFT	US Stock	Binary (t=	1.260	0.998	0.262	27.6
MUB	Bond ETF	Adaptive	0.769	0.313	0.456	6.5
Maaden	TASI Stock	Binary (t=	0.824	0.505	0.319	105.3
Mobily	TASI Stock	Binary (t=	1.264	0.728	0.536	29.0
NFLX	US Stock	Scaled	1.162	0.729	0.433	54.9
NVDA	US Stock	Binary (t=	1.374	1.251	0.123	59.3
NVO	Intl Stock	Scaled	0.910	0.495	0.415	30.5
PALL	Commodity	Binary (t=	0.640	0.232	0.408	36.4
PEP	US Stock	Scaled	1.039	0.546	0.493	18.6
PG	US Stock	Scaled	1.056	0.629	0.427	20.1
PPLT	Commodity	Binary (t=	1.153	0.313	0.840	26.4
QQQ	Broad ETF	Adaptive	1.281	1.026	0.255	24.2
REM	REIT	Adaptive	0.437	0.238	0.199	29.4
RIO	Intl Stock	Scaled	0.738	0.438	0.300	38.7
Riyad_Bank	TASI Stock	Binary (t=	0.951	0.468	0.483	27.6
SABIC	TASI Stock	Binary (t=	0.440	-0.105	0.545	24.1
SADAFCO	TASI Stock	Binary (t=	0.607	0.436	0.171	25.0
SAP	Intl Stock	Binary (t=	0.833	0.548	0.285	35.5
SHEL	Intl Stock	Binary (t=	0.835	0.430	0.405	26.9
SHY	Bond ETF	Adaptive	1.388	1.070	0.318	1.4
SLV	Commodity	Binary (t=	0.748	0.510	0.238	32.5
SPY	Broad ETF	Adaptive	1.042	0.890	0.152	17.7
STC	TASI Stock	Scaled	0.813	0.576	0.237	60.5
Saudi_Kayan	TASI Stock	Binary (t=	0.702	-0.072	0.774	31.7
TAN	Thematic ETF	Adaptive	0.699	0.378	0.321	44.6
TIP	Bond ETF	Scaled	0.901	0.454	0.447	5.9
TLT	Bond ETF	Scaled	0.953	0.119	0.834	13.4
TM	Intl Stock	Scaled	1.166	0.517	0.649	25.0

Asset	Class	Best Strat	Best S	BnH S	Excess	Vol%
TSLA	US Stock	Binary (t=	1.050	0.816	0.234	55.4
TSM	Intl Stock	Binary (t=	1.326	1.029	0.297	37.9
Tawuniya	TASI Stock	Binary (t=	1.119	0.297	0.822	97.3
UNG	Commodity	Scaled	0.603	-0.201	0.804	48.2
UNH	US Stock	Binary (t=	0.962	0.706	0.256	32.3
USO	Commodity	Binary (t=	0.551	0.232	0.319	37.1
UUP	Currency	Binary (t=	0.827	0.447	0.380	7.6
V	US Stock	Binary (t=	1.050	0.873	0.177	24.9
VNQ	REIT	Binary (t=	0.803	0.368	0.435	25.0
VNQI	REIT	Binary (t=	0.562	0.065	0.497	16.9
VWO	Intl ETF	Scaled	0.688	0.381	0.307	22.3
WMT	US Stock	Binary (t=	0.936	0.790	0.146	22.8
XBI	Thematic ETF	Binary (t=	0.827	0.333	0.494	29.1
XLB	Sector ETF	Binary (t=	0.904	0.560	0.344	22.4
XLE	Sector ETF	Binary (t=	0.682	0.435	0.247	26.8
XLF	Sector ETF	Binary (t=	0.908	0.727	0.181	26.1
XLI	Sector ETF	Binary (t=	1.094	0.746	0.348	20.7
XLK	Sector ETF	Binary (t=	1.222	1.043	0.179	23.6
XLP	Sector ETF	Binary (t=	1.145	0.784	0.361	13.9
XLRE	Sector ETF	Adaptive	0.779	0.411	0.368	20.8
XLU	Sector ETF	Binary (t=	0.965	0.620	0.345	18.2
XLV	Sector ETF	Scaled	1.201	0.877	0.324	17.3